Database Introduction

1. Define database and database management system (DBMS). Write down the advantages of database management system over traditional file processing system.
2. What is Data abstraction? Explain in brief different level in data abstraction.
3. Explain functional component of database management system.
4. What do you mean by Database Administrator?The responsibilities /function of a database administrator.
5. What do you mean by relational database. Write down the advantages of relational database.
6. Define DDL and DML. What are the Difference between procedural and nonprocedural DML.
7. What is an E-R diagram? construct an E-R diagram for a car insurance company. Consider each customer owns a number of cars and each car is associated with a number of recorded accident.
8. What do you mean by physical and logical data independence.
9. What is data model? Describe relational Data Model with example.
10. What is databaseschema.Difference between instances and database schema.
11. Explain storage management.
12. Describe Database System Structure with fig.
13. Describe different types of database system user.
14. Explain the component of query processor.
15. Briefly describe the two-Tier and Three- Tier architecture of database management system.

Chapter-2

Entity-Relationship Model

1. Explain the entity relationship model.
2. Define entity, super key, attributes, candidate key and primary key with appropriate example.
3. What do you mean by Entity, Entity set and Attributes?
4. What is mapping cardinalities? Describe various mapping cardinalities.

5. What do you mean by simple, single-Valued, Multivalued, Null and Derived attributes?
6. Differentiate among super key, primary key, and candidate key with example.
7. Define foreign key. Why it is used.
8. What is the component of an E-R diagram?
9. Explain strong entity set & weak entity set. Explain difference between strong entity set and weak entity set.
10. Explain Specialization and Generalization. What are the difference between Specialization and Generalization?
11. Explain aggregation. Why it is used in E-R modeling.
12. What do you mean by Attribute Inheritance?
13. Define degree and tuple.

Chapter-6

1. What dou you mean by transaction? Show and explainthe state diagram of a transaction.
2. Describe the ACID properties of transactions.
3. Define local transaction and global transaction.
4. Explain consistency, atomicity and durability.
5. Explain the usefulness of concurrent execution.
7. Define serializability. Explain the conflict and view serializability.

Chapter-7
Integrity Constraints And Relational Database Design

1. What do you mean by Domain Constraints? What are Assertions?
2. What do you mean by Triggers?. What do you mean by Functional Dependency?Explain full and partial functional dependencies with example.
3. Explain the ways of using functional dependences.
4. Write down the rules of Functional Dependency.
5. What do you mean by normalization? Explain different forms of Normal Form.
6. Explain Boyee-Code Normal Form(BCNF) with example.

7. Explain multivalued dependency (MVD).
8. Explain not null constraint, unique constraint.
10. Describe referenTal integrity.
11. What do you mean by transitive dependency?
12. Explain superkey in terms of functional dependency.
13. Explain trivial and nontrivial dependencies.
14. Explain join dependencies.
15. Show that join dependency is not equivalent to multi-valued dependency.
16. Normalize with explanation the following table into 1NF, 2NF and 3NF :- PROJECT (Proj-ID, Proj-Name, Pro-Mgr-ID, Emp-ID, Emp-name, Emp-Dept, Emp-Hrly-rate, Total-Hrs)
17. Normalize with explanation the following table into 1NF, 2NF and 3NF :- Emp-proj (SSN, Pnumber, hours, Ename, Pname, Ddate Address, Pnumber, Plocation, Dnames, DMGss)
18. Explain the difference between BCNF and 3NF.
19. Consider the following set of functional dependencies on schema (A,B,C):- A -› BC B -› C A -› B AB-› C Calculate the canonical cover for F. (2009)
20. Describe the Armstrong's rules.
21. Compute the closure of the following set F of Functional dependencies for relation schema: (2010,2011) R = (A, B, C, D, E, F) A -› BC CD -› E B -› D E -› A 23. Explain the desirable properties of decomposition. (2011)
24. What do you mean by lossless and lossy decompositions? Give examples.

Chapter-12

Indexing and Hashing

1. What is indexing and hashing?
2. Why indexing is used in database?
3. Define primary and secondary indices.
4. What are the difference between primary and secondary index?
5. Explain the types of indices. 6. Explain why dense indices are preferable.
7. Write short notes on B+ -Tree index files.
8. Write short notes on Hash file organization.
9. Difference between B Tree and B+ -tree indexing.
10. Define multilevel indexing. When it is preferable to use a multilevel index rather than a single-level index? Explain.

11. What are the limitation of order indexing over direct indexing technique?
12. What are the difference between dense and spares indexing.